

English Roadmap

My Complete End-to-End Machine Learning Pipeline Study Summary (Scikit-Learn)

From raw data → feature engineering → preprocessing → dimensionality reduction → modeling → evaluation → model persistence

Over the past weeks, I've completed a systematic review of the **entire machine learning workflow using scikit-learn**, and I'm summarizing my notes here. This is the condensed version of my full study document — focusing on concepts, when to use each technique, and practical decision rules.

1 Data Loading

Scikit-learn provides several built-in datasets for quick experimentation:

- `load_iris`, `load_wine`, `load_digits`, `load_breast_cancer`
- For real projects: `pandas.read_csv()` or database connectors

Checklist before starting a project:

- Missing values?
 - Mixed data types (numerical + categorical)?
 - Outliers?
 - Imbalanced target?
-

2 Feature Engineering (Turning data into ML-ready features)

2.1 Categorical Features

Technique	When to Use	Notes
OneHotEncoder	Few stable categories	Produces binary columns

Technique	When to Use	Notes
DictVectorizer	Input is list of Python dicts	Similar to One-Hot
CountVectorizer / TF-IDF	Text data	Bag-of-words / TF-IDF
LabelEncoder	<i>Only for y (labels)</i>	Never for feature columns

Rule:

➡ Fit encoders on **train only**, then transform **test**.

2.2 Numerical Features

- **PolynomialFeatures** → Add interactions / non-linear terms
- **Binning / discretization** → Good for skewed distributions

3 Preprocessing (Cleaning + Scaling)

3.1 Missing Values

- `SimpleImputer(strategy="mean")` → numeric
- `median` → robust to outliers
- `most_frequent` → categorical
- `KNNImputer` → best performance but slow

3.2 Scaling

Scaler	When to Use
StandardScaler	Most common; required for logistic regression, SVM, KNN, PCA, KMeans
MinMaxScaler	Fixed feature range; images
RobustScaler	Heavy outliers

➡ Many ML algorithms are **scale-sensitive**.

4 Dimensionality Reduction / Feature Selection

4.1 PCA (Principal Component Analysis)

- Unsupervised
- Maximizes captured variance
- Essential for high-dimensional or noisy features
- Requires StandardScaler

4.2 LDA (Linear Discriminant Analysis)

- Supervised
- Maximizes class separation
- Max components = $n_{\text{classes}} - 1$
- Good for classification visualization / compression

4.3 Feature Selection

Method	Use Case
VarianceThreshold	Remove near-constant features
SelectKBest (chi2 / f_classif / mutual_info)	Filter-based selection
RFE / model-based selection	When interpretability matters

5 Modeling (Classification & Regression)

5.1 Classification Algorithms

Algorithm	Strengths	Notes
Logistic Regression	Fast, interpretable, strong baseline	Needs scaling
KNN	Simple, flexible	Expensive prediction; scale-sensitive
Decision Tree	Interpretable; no scaling needed	Can overfit

Algorithm	Strengths	Notes
Random Forest	Powerful default model	Works extremely well on tabular data
Naive Bayes	Ideal for text	Very fast

5.2 Regression Algorithms

Algorithm	Highlights
Linear Regression	Baseline
Ridge (L2)	Stabilizes training, good for high-dim data
Lasso (L1)	Feature selection (weights $\rightarrow 0$)
ElasticNet	Balanced L1+L2
SGDRegressor	Handles extremely large datasets

6 Model Evaluation

- Classification: **accuracy, F1, precision/recall, ROC-AUC**
- Regression: **MAE, MSE, RMSE, R^2**
- Cross-validation: `cross_val_score`, `GridSearchCV`, `RandomizedSearchCV`

7 Saving & Loading Models

Using joblib:

```
import joblib

joblib.dump(model, "best_model.pkl")
model = joblib.load("best_model.pkl")
```

Supports saving:

- Models
- Pipelines
- Preprocessing steps (scalers, encoders)

8 Choosing the Right Tool (Cheat Table)

8.1 Feature Engineering

Data Type	Best Method
Categorical	OneHotEncoder
High-cardinality text	TF-IDF
Dictionary-like	DictVectorizer
Numerical with skew	Binning

8.2 Preprocessing

Problem	Tool
Missing numbers	SimpleImputer
Scale imbalance	StandardScaler
Heavy outliers	RobustScaler
Non-linear patterns	PolynomialFeatures

8.3 Reduction / Selection

Scenario	Method
Very high-dimensional	PCA
Classification	LDA
Remove weak features	SelectKBest

8.4 Model Choice

Scenario	Best Algorithm
General tabular data	Random Forest
High-dim sparse text	Logistic Regression / Linear SVM
Non-linear small dataset	KNN
Need feature importance	Decision Tree / Random Forest
Need speed	Naive Baye

